



Module 11 Day 2: Cryptography

Solution Guide: GPG

In this activity, you practiced using GPG for a real-world application of public key cryptography, generating public and private keys and encrypting and decrypting secret messages with the appropriate keys.

This solution is an example of how to encrypt and decrypt with GPG.

- First come up with an idea of how to capture an insider at the Hill Valley Police Department, and add it to a message in a file called `secret_idea.txt`.
 - You can use Vi, Nano, or echo to create this message.
- The next step is to generate your public key:
 - `gpg --gen-key`
- Validate your key has been generated:
 - `gpg --list-keys`
- Export the public key in ASCII format:
 - `gpg --armor --output [your name]_key.gpg --export [your email]`
 - Update the command with your name and the email you used for generating the key.
- Confirm the key has been generated by running:
 - `more [your name]_key.gpg`
- It should look similar to the following:
Note: This is a shortened example.

Unset

```
-----BEGIN PGP PUBLIC KEY BLOCK-----
```

```
oCN2AghQUdgu5yBVAmPAx7hatvcMBR1X6NqJN4wStLB210vHdgT2VbiHUtw  
kGvbJ
```

```
Hsui9eTR7bBY1YgP8PcGFjeMZ5+C7E94uYeksbwMzFWGE79M3kqEi1tgkDZ
TN/T8
```

```
8031qQUgDCCbUnuwpW5pYJ2BconeNBHAZNKSKg+9U3DfCazRpky89be6W7W
tjDGs
```

```
iFo5PEjBTvCJJXHvDgn2W7I7U0MW0220gyCT/Ja/eKad5GKTeMj0C4ERTwv
ha00N
```

```
-----END PGP PUBLIC KEY BLOCK-----
```

- Send your key to your partner via Slack.
 - You can send the whole file or copy and paste the message (including the -----BEGIN PGP PUBLIC KEY BLOCK----- and the -----END PGP PUBLIC KEY BLOCK-----).
 - If you receive text from your partner instead of a file, create a file called `partner_key.gpg`, add the whole message, and save the file.
- With your partner's public key, import by running the following:
 - `gpg --import [partner_key.gpg]`
- Encrypt your message with your partner's public key by running the following command:
 - `gpg --output [your name]_secret_idea.txt.enc --encrypt --recipient [partner key] secret_idea.txt`
- Send the encrypted file `[your name]_secret_idea.txt.enc` to your partner.
- When you receive your partner's encrypted file, save it in your local directory.
- Decrypt the encrypted message with the following command:
 - `gpg --output decrypted_message.txt --decrypt [your partner's name]_secret_idea.txt.enc`
- Preview the decrypted message and confirm with your partner this is the message they sent.